

Quantum Field Theory and Statistical Systems

Casimir-driven traversable wormholes: Geometry, stability and observable signatures

 Madhukrishna Chakraborty ^a, Subenoy Chakraborty ^{b,c,d,*}
^a Department of Mathematics, Techno India University, Kolkata, 700091, West Bengal, India^b Department of Mathematics, Brainware University, 700125, Kolkata, West Bengal, India^c Shinawatra University, 99 Moo 10, 12160, Bangtoey Pathum Thani, Samkhok, Thailand^d INTI International University, Persiaran Perdana BBN Putra Nilai, 71800, Nilai, Malaysia

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ABSTRACT

The paper deals with the study of some novel traversable Wormholes (TWHs) in the presence of Casimir effect, a well known quantum phenomenon in conjunction with classical electric charge and a massless scalar field with vanishing potential. The paper delves deep into exploring the interplay of General Relativity (GR) and Quantum Field Theory (QFT) to shape realistic models of Wormhole (WH) space-times. Further, it studies the geometry and stability of these WHs both on qualitative as well as on quantitative grounds. Finally, the paper provides valuable insights regarding the observational signatures like shadows, quasinormal modes (QNMs) and grey body factors (GBFs) concerning the TWHs so constructed.

1. Introduction

Wormholes (WHs) are hypothetical compact objects arising from solutions to Einstein's field equations, serving as theoretical bridges connecting distant regions within the same universe or entirely separate universes. The concept dates back to the 1935 work of Einstein and Rosen, who proposed a geometric model—the Einstein-Rosen bridge—that linked two asymptotically flat regions. However, this bridge was not traversable. It was later, through the work of Morris and Thorne, that the possibility of traversable wormholes (TWHs) gained attention. They showed that maintaining such a structure would require exotic matter with negative pressure, which leads to a violation of classical energy conditions [1–5]. Thus, researchers have searched for the WH configuration in Einstein gravity as well as in a plethora of extended theories of gravity generating effective exotic fluid satisfying one or more classical energy conditions (ECs) like higher order gravity and string theory [6], Einstein Gauss Bonnet Gravity theory [7], $f(Q)$ gravity [8], LTB [9], Horava-Lifschitz gravity [10], dRGT massive gravity [11], $f(R)$ gravity [12], teleparallel $f(T)$ gravity [13], scalar-tensor theories [14] etc.

On the other hand, dark energy motivated exotic matter with standard Einstein-Hilbert action has also gained interest. Some of them are reported in [15–17]. Therefore, it is well understood that both extended theories of gravity and some forms of exotic matter can stabilize the WH throat. However, these models are purely theoretical and hence suffer from some sort of inconsistencies. These are not tested or verified via experiments in laboratories. However, a special type of vacuum energy known as Casimir energy has been confirmed by the experiment. Also, it is believed to be a successful candidate of matter forms used to stabilize the TWHs. Additionally, the classical electric and magnetic fields help to assist stabilization of the WHs. A study of static charged WHs can be

* Corresponding author.

E-mail addresses: madhukrishna.c@technoindiaeducation.com (M. Chakraborty), schakraborty.math@gmail.com (S. Chakraborty).

found in [18–20]. Charges play role in stabilization of WH throat, they are believed to be a natural extension of the Morris-Thorne type TWHs.

Since its inception [21,22], in the recent times, there has been development in the Casimir WH [23–25]. For stabilization of WH throat, violation of Null Energy Condition (NEC) is guaranteed by some forms of exotic matter and one such practical example can be found in Casimir effect of QFT [26] where one can get into the infinite vacuum energy just by placing two parallel conducting plates. In 1996, Lamoreaux experimentally confirmed the assertion of this effect [27]. Later, by improving the experimental set up, Bressi [28] obtained a better result. Results corresponding to the Casimir effect for parallel plates are well established in literature. It is a bit difficult to calculate the Casimir effect when the shape of the conductor is varying. There are two popular approximation methods viz. Proximity Force Approximation (PFA) and the multiple scattering approach which are well known in the literature [29].

In the present work, we aim to find some exact novel solutions for TWHs in the presence of variable charged Casimir plates to find the effect of charge and casimir energy in geometry and stability of the TWHs. The paper also shows the analytical study of these WH shadows. This is because, after the detection of Black-hole (BH) shadows of $M87^*$ and $SgrA^*$, researchers are interested to study the WH shadows likewise. Shadows of a variety of compact objects using Event Horizon Telescope (EHT) datasets those have been studied so far can be found in [30,31]. Wormhole and black hole shadows are fundamentally different due to their structure but they are crucial concepts exhibiting distinct observational features. A black hole shadow is casted due to the gravitational bending of light around an event horizon, the boundary beyond which nothing, not even light, can escape also known as a point of no return. A black hole shadow (for a supermassive black hole, such as $SgrA^*$ or $M87^*$) is basically determined by its size, shape, event horizon and Photon ring structure. For $M87^*$ and $SgrA^*$ (observed by the Event Horizon Telescope (EHT)), both shadows appear roughly circular but with minute features that depend on the shape, the spin of the photon ring, accretion and the surrounding plasma dynamics. Unlike black holes, wormholes do not have an event horizon in the usual sense. Instead, they possess a passage that allows light to escape from one side of the wormhole to the other. Thus, the shadow might not exhibit the clear boundary associated with a black hole's event horizon but could instead display more insightful features based on the geometry of the WH. If a WH is traversable, there may be light escaping from the opposite side of the WH (depending on its specific geometry). This could lead to more complicated emission spectra around the shadow, potentially observable in high-resolution images or time-varying data (e.g., changes in light intensity as material passes through or around the wormhole). On the observational grounds, distinguishing a black hole and a wormhole shadow is not so easy, especially with current technology. The Event Horizon Telescope (EHT), which captured the first images of the shadows of $M87^*$ and $SgrA^*$, has resolution limitations. The shadow of a black hole has been detected but there is at present no way to definitely rule out the possibility of WHs purely from observational data. Motivated by the EHT papers on shadow [31–35] in the present Casimir-driven TWHs, we analyze the effect of Casimir energy, electric charge and massless scalar field with vanishing potential in the shape and size of the shadows.

On the other hand, QNMs and GBFs are very promising areas of gravity wave astronomy. They depict how the perturbations such as Gravity waves behave near any compact astrophysical object. In this context, it is worthy to mention that a nice interplay between QNMs and GBFs has been established in [36,37]. GBFs describe how incoming radiation from BH is partially transmitted and partially reflected due to the curvature of space-time. GBFs may exhibit higher stability against deformations of the near horizon geometry [38] when compared to the overtones of QNMs that are highly sensitive to such deformations. The correspondence between QNMs and GBFs for BHs have been found using WKB approach. Given that, the boundary conditions for QNMs and GBFs are the same for BHs and WHs, it is reasonable to consider such similar correspondence in case of WHs as well. In fact, in terms of tortoise coordinate r_T , we can distinguish WH from a BH. In case of BH, $-\infty < r_T < +\infty$ represents the event horizon and only one side of the asymptotic region, while $-\infty < r_T < +\infty$ takes into account the two asymptotically flat space-time regions of a WH without any horizon. Although, the literature on QNMs of WHs is quite vivid, the calculation of GBFs is model-specific.

Casimir WHs are the hypothetical objects supported by Casimir effect that provides exotic matter needed for the possible stabilization of the WH throat, allowing researchers to explore QFT in curved space-time. Analysis on shadows of TWHs provides observational evidence of GR and exotic space-time geometries. QNMs describe the fingerprint of the geometry allowing precise characterization of the compact object. For Casimir WHs, the QNM spectra could differ from that of a BH probing the stability of the WH under perturbations. Grey body factors give idea of how the potential barrier around a compact object affects the escape/ transmission of radiation. GBFs for Casimir WHs might give us some insights about the quantum effects in semi-classical settings. QNMs, GBFs and shadows altogether offer valuable insights for information transfer through wormholes and whether quantum vacuum stress-energy supports this traversable geometry. Casimir effect is a well-known quantum phenomenon. The paper is aimed at exploring the interplay of GR, QFT and finding answer to the question ‘Can quantum vacuum effects like the Casimir effect provide the required negative energy density to support traversability?’, if yes then this would enable us to investigate the fundamental question of information travel and causality: one of the finest takeaways of studying the WH geometry.

The layout of the paper is as follows: Section 2 deals with a brief overview of Einstein's field equations together with Casimir effect and WH configuration. Section 3 deals with the TWH solutions, their traversability and ECs. Section 4 deals with the embedding diagrams and focuses on the quantitative analysis of the obtained WH solutions. Section 5 deals with the analysis of the WH shadows. Section 6 gives the detailed analysis of QNM spectra and GBF for the obtained TWHs. The paper ends with concluding remarks in Section 7.

2. Overview of TWHs and Casimir effect with massless scalar field coupled with electric charge

The line element representing a spherically symmetric static WH space-time is given by [39]

$$ds^2 = -e^{2\Phi(r)} dt^2 + \frac{1}{\left(1 - \frac{b(r)}{r}\right)} dr^2 + r^2 d\Omega_2^2 \quad (1)$$

where $d\Omega_2^2 = d\theta^2 + \sin^2\theta d\phi^2$, $b(r)$ is called the WH shape function which satisfies the metric condition $\left(1 - \frac{b(r)}{r}\right) > 0$. The throat condition is given by $b(r_0) = r_0$, where r_0 is the radius of the throat. In order to have a TWH structure, flaring out condition given by $rb'(r) - b(r) < 0$ must hold where prime ' denotes differentiation w.r.t r , the radial coordinate. $\Phi(r)$ is called the red-shift function. The radial coordinate r ranges from r_0 i.e., the throat radius to infinity. $\Phi(r)$ is finite everywhere to avoid singularity or horizon and is used for the detection of signal coming from a distant observer. It also encodes information about the radial tidal force. In order to have asymptotic flatness, $b(r)$ should satisfy $\frac{b(r)}{r} \rightarrow 0$ as $r \rightarrow \infty$.

In GR, the following are the classical ECs that play a crucial role in the formation of WHs [40].

1. Weak Energy Condition (WEC): $\rho \geq 0$, $\rho + p_r \geq 0$, $\rho + p_t \geq 0$.
2. Null Energy Condition (NEC): $\rho + p_r \geq 0$, $\rho + p_t \geq 0$.
3. Dominant Energy Condition (DEC): $\rho - |p_r| \geq 0$, $\rho - |p_t| \geq 0$.
4. Strong Energy Condition (SEC): $\rho + p_r \geq 0$, $\rho + p_t \geq 0$, $\rho + p_r + 2p_t \geq 0$.

Using Raychaudhuri equation (RE), one can show that in order to satisfy the flaring out condition near the throat the matter at the throat should violate NEC necessitating some form of exotic matter which although may be theoretically stated but is quite difficult to harness in reality. However, there are certain physical phenomena where such violations occur like the Casimir experiment [23–25] where two conducting plates are placed close to each other so that they distort the zero-point energy leading to a difference in pressures between inner and outer walls. Consequently, the plates attract each other. This is because, there are fewer virtual particles per unit volume between them than in the space outside the plates, indicating the presence of negative energy density in the gap. The Casimir effect gives an experimental evidence of the violation of ECs by quantum effects. Within this framework, attempts have been made to utilize the Casimir source as a form of exotic matter to stabilize the WH throat, leading to Casimir WHs [23–25]. There are cases where the Casimir source is accompanied by electric field generated by a point-charge acting as a source of positive energy density.

The Casimir source can be described by a Stress-Energy Tensor (SET) of the form [41]

$$T_\mu^\nu|_{\text{Casimir}} = \text{diag}[-\rho_C(r), p_{r,C}(r), p_{t,C}(r), p_{t,C}(r)] = \frac{-\hbar c \pi^2}{720r^4} \text{diag}[-1, 3, -1, -1] \quad (2)$$

while an electric field has a SET of the form [41]

$$T_\mu^\nu|_{\text{Electric field}} = \text{diag}[-\rho_E(r), p_{r,E}(r), p_{t,E}(r), p_{t,E}(r)] = \frac{Q_0^2}{32\pi^2\epsilon_0 r^4} \text{diag}[-1, -1, 1, 1] \quad (3)$$

The components of the electric charge density and the radial pressure obey the following equation of state (EoS)

$$\rho_E(r) = -p_{r,E}(r) \quad (4)$$

Consequently, NEC of the combined source is violated as

$$\rho(r) + p_r(r) = \rho_C(r) + p_{r,C}(r) + \rho_E(r) + p_{r,E}(r) = -\frac{\hbar c \pi^2}{180r^4} < 0 \quad (5)$$

Therefore, introduction of electric charge does not alter the violation of the NEC by means of the Casimir source. However, by replacing electric field by a scalar field we may alter the violation of NEC. It is well accepted that scalar field having opposite polarity commonly referred as phantom fluid can serve as a source for TWHs without any Casimir source. One may refer to [41] for the description of exotic matter with scalar field (both with and without mass). C. Barcelo and M. Visser [42] examined the effect of a pure scalar field conformally coupled with a gravitational field. On the other hand L. Butcher [43] proved that every form of pure scalar field can not be taken as a source for a TWH. While [44] studies charged Casimir wormholes in the framework of the Generalized Uncertainty Principle, our work differs in both the underlying assumptions and the scope of the analysis. This work lies entirely within the standard (non-GUP) Einstein gravity framework, with the Casimir term introduced directly into the stress-energy tensor alongside a massless scalar field and electromagnetic field. This allows us to isolate and separately analyze the effects of each source on the wormhole geometry, without the additional GUP-induced modifications to the field equations. In addition to the standard geometric analysis, we compute shadows, quasinormal modes, and grey body factors for each model, providing a broad set of observational signatures. We treat the Casimir parameter r_1 as a variable quantity to explore qualitative sensitivity, whereas in the GUP framework the Casimir contribution is tied to the deformation parameter. This separation allows a cleaner understanding of the Casimir effect in a standard GR setting. We consider the SET of the form

$$T_\mu^\nu = \text{diag}[-\rho(r), p_r(r), p_t(r), p_t(r)] \quad (6)$$

where $\rho(r)$ represents the total energy density, $p_r(r)$ represents the radial pressure and $p_t(r)$ stands for the tangential pressure of the anisotropic SET. The conservation of energy-momentum gives the following differential equation

$$p'_r(r) = \frac{2}{r} (p_t(r) - p_r(r)) - (\rho(r) + p_r(r))\Phi'(r) \quad (7)$$

We consider a SET comprising of $T_{\mu\nu}^{scalar}$ originating from the scalar field, $T_{\mu\nu}^{Casimir}$ resulting from the Casimir source and the last part coming from

$$T_{\mu}^{\nu}|_{THW} = \text{diag}[-\rho^{TWH}(r), p_r^{TWH}(r), p_t^{TWH}(r), p_t^{TWH}(r)] \quad (8)$$

The action governing this system is given by [41]

$$S = \int d^4x \sqrt{-g} \left(\frac{R}{2\kappa} - \frac{1}{2}(\nabla\psi)^2 - V(\psi) + \mathcal{L}_C \right) \quad (9)$$

where R is the Ricci scalar, $g = \det g_{\mu\nu}$, $g_{\mu\nu}$ is the metric tensor, $\kappa = 8\pi G$, ψ is the scalar field, $\mathcal{L}_C$ is the Lagrangian of the Casimir source and $V(\psi)$ is the potential of the scalar field such that $V(\psi) = V[\psi(r)] = V(r)$ if we impose $\psi = \psi(r)$. Varying the action (9) w.r.t the scalar field we get

$$\nabla^2\psi = \frac{dV(\psi)}{d\psi} \quad (10)$$

In the present analysis, we consider $V(\psi) = 0$ that corresponds to the massless case. This implies, $\nabla^2\psi = 0$ or

$$e^{2\Phi(r)} r^4 \left(1 - \frac{b(r)}{r} \right) \psi'(r)^2 = C \quad (11)$$

where $C > 0$ is a constant. On the other hand, varying the action (9) w.r.t the metric we get the field equations as follows [41]

$$\frac{b'(r)}{r^2} = \kappa\rho(r) \quad (12)$$

$$\frac{2\Phi'(r)}{r} \left(1 - \frac{b(r)}{r} \right) - \frac{b(r)}{r^3} = \kappa p_r(r) \quad (13)$$

$$\left(1 - \frac{b(r)}{r} \right) \left(\Phi''(r) + \Phi'(r)^2 + \frac{\Phi'(r)}{r} \right) + \frac{(b(r) - rb'(r))}{2r^3} (r\Phi'(r) + 1) = \kappa p_t(r) \quad (14)$$

The expressions for ρ , p_r , p_t are [41]

$$\rho = \frac{C e^{-2\Phi(r)}}{r^4} - \frac{r_1^2}{\kappa r^4} + \frac{Q_0^2}{r^4} \quad (15)$$

$$p_r = \frac{C e^{-2\Phi(r)}}{r^4} - \frac{3r_1^2}{\kappa r^4} - \frac{Q_0^2}{r^4} \quad (16)$$

$$p_t = -\frac{C e^{-2\Phi(r)}}{r^4} + \frac{r_1^2}{\kappa r^4} + \frac{Q_0^2}{r^4} \quad (17)$$

The first term in each of the above equations is due to massless scalar field having zero potential. The second term comes from the Casimir source. The third term comes from electromagnetic field. This shows that, we have considered variable Casimir plates with electric charge and massless scalar field with vanishing potential to construct TWHs. The Casimir WHs are expected to be distinct from the other WH solutions as here we have explicit expressions for the total energy density, radial and tangential pressure components. So, we don't assume any linear relation among (ρ, p_r) or (p_r, p_t) . The parameter r_1 arises as an integration constant associated with the Casimir source term in the field equations. In a fully specified physical setup, this scale would be determined uniquely by the chosen boundary conditions and renormalization scheme. In the present work, we treat r_1 as a variable parameter solely to investigate how changes in the Casimir contribution affect the wormhole geometry and related observables.

3. Casimir WHs: Traversability and energy conditions

3.1. WH solution 1

Taking derivative of Eq. (16) and equalizing it with Eq. (7) we get

$$e^{\Phi(r)} = \Phi_0 \left(\frac{r}{r_0} \right) \quad (18)$$

where Φ_0 is a constant of integration. We have used the expressions for ρ , p_r and p_t from Eqs. (15)–(17). Using field Eq. (12), the expression for ρ from Eq. (15) and the expression for $\Phi(r)$ from (18) we get the WH shape function as

$$b(r) = b_0 - \frac{\kappa C r_0^2}{3\Phi_0^2 r^3} + \frac{r_1^2}{r} - \frac{\kappa Q_0^2}{r} \quad (19)$$

Clearly, $\frac{b(r)}{r} \rightarrow 0$ as $r \rightarrow \infty$ thereby assuring the asymptotic flatness and flaring out holds as shown in Fig. 1.

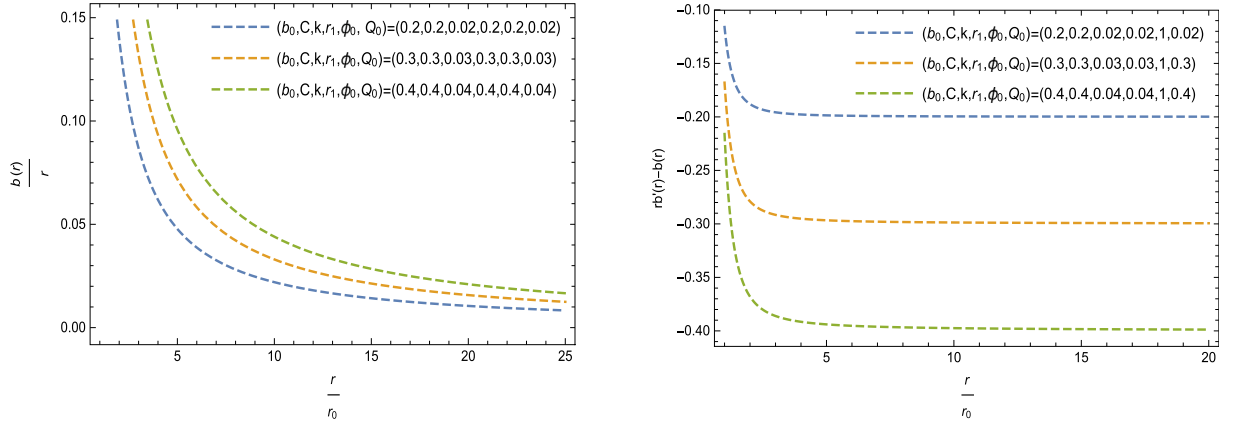


Fig. 1. Asymptotic flatness (left) and flaring-out condition (right) for WH1.

The expressions for the ECs can be written as

$$\rho(r) = \frac{Cr_0^2}{\Phi_0^2 r^6} - \frac{r_1^2}{\kappa r^4} + \frac{Q_0^2}{r^4} \quad (20)$$

$$\rho + p_r + 2p_t = \frac{2Q_0^2}{r^4} - \frac{2r_1^2}{\kappa r^4} \quad (21)$$

$$\rho + p_r = \frac{2Cr_0^2}{\Phi_0^2 r^6} - \frac{4r_1^2}{\kappa r^4} \quad (22)$$

$$\rho + p_t = \frac{2Q_0^2}{r^4} \quad (23)$$

$$\rho - p_r = \frac{2r_1^2}{\kappa r^4} + \frac{2Q_0^2}{r^4} \quad (24)$$

$$\rho - p_t = \frac{2Cr_0^2}{\Phi_0^2 r^6} - \frac{2r_1^2}{\kappa r^4} \quad (25)$$

It is observed that with Casimir effect, massless scalar field with vanishing potential and electric field it is possible to get asymptotic flat TWH satisfying flaring out criteria and the combined effects characterize a WH configuration where classical ECs may hold if the effect of massless scalar field with zero potential coming via C and the electromagnetic field via Q_0 both are dominant over the Casimir source via r_1 .

Note: Indeed, with the current choice $\Phi(r) \propto \ln(r)$, the lapse function does not approach a constant at infinity, so the metric is not strictly asymptotically flat although the text uses $\frac{b(r)}{r} \rightarrow 0$ as $r \rightarrow \infty$ to define asymptotic flatness and this holds for WH1. However, in the present work, our main interest lies in the local and near-throat properties of the WH geometry and their phenomenological consequences. All physical quantities such as lensing observables, quasinormal modes, and stability indicators those have been computed depend only on the geometry up to a large but finite radius say $r = R$, much before the region where the asymptotic deviation would largely dominate. For such cases, it is fully consistent to match the WH1 solution at $r = R$ to an exterior, asymptotically-flat vacuum region (e.g., Minkowski or Schwarzschild) via standard Darmois-Israel junction conditions. This procedure restores global consistency while keeping all local results unchanged.

3.2. WH solution 2

For this particular case, we consider $\Phi = \Phi_0$, a constant. Then using this in Eqs. (13) and (16) we obtain the shape function as

$$b(r) = \frac{-\kappa C e^{-2\Phi_0}}{r} + \frac{3r_1^2}{r} + \frac{\kappa Q_0^2}{r} \quad (26)$$

Consequently, the radius of the throat is given by

$$r_0 = \sqrt{3r_1^2 + \kappa Q_0^2 - \kappa C e^{-2\Phi_0}} \quad (27)$$

The asymptotic flatness and flaring-out criteria for WH2 has been depicted graphically in Fig. 2.

The expressions for the ECs in case of WH2 can be written as

$$\rho = \frac{C e^{-2\Phi_0}}{r^4} - \frac{3r_1^2}{\kappa r^4} - \frac{Q_0^2}{r^4} \quad (28)$$

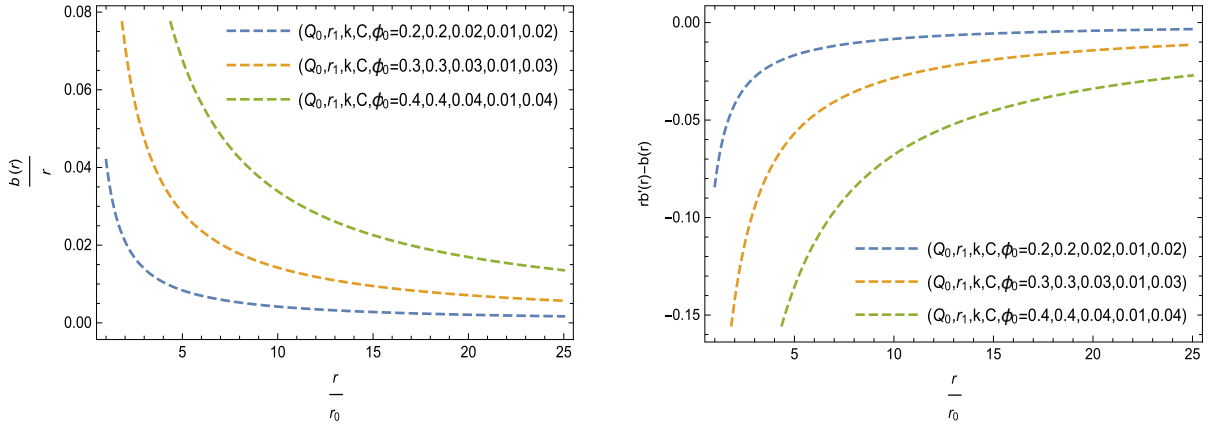


Fig. 2. Asymptotic flatness (left) and flaring-out condition (right) for WH2.

$$\rho + p_r = 2 \frac{C e^{-2\Phi_0}}{r^4} - \frac{6r_1^2}{\kappa r^4} \quad (29)$$

$$\rho + p_t = 0 \quad (30)$$

$$\rho - p_r = -\frac{2Q_0^2}{r^4} \quad (31)$$

$$\rho - p_t = 2 \left(\frac{C e^{-2\Phi_0}}{r^4} - \frac{3r_1^2}{\kappa r^4} - \frac{Q_0^2}{r^4} \right) \quad (32)$$

$$\rho + p_r + 2p_t = \frac{2}{r^4} Q_0^2 \quad (33)$$

For WH2, the SEC holds as seen from Eq. (33), DEC is violated but NEC holds if the effect of Casimir energy is minimal via r_1 and the effect of massless scalar field via C is dominant.

Note: Although the flare-out and asymptotic flatness conditions are purely geometric and do not directly depend on the redshift function, in our plots the parameter Φ_0 appears in $b(r)$ because the full analytical solution of the Einstein field equations couples $\Phi(r)$ and $b(r)$. Its presence is therefore a consequence of the solution's structure, not an independent influence $\Phi(r)$ on the geometric criteria.

Remark on the role of the scalar field: In the present multi-source WH models, the massless scalar field is not introduced arbitrarily but as an essential component of the total stress-energy tensor, alongside the electromagnetic field and the Casimir term. Its contribution allows for analytic integration of the Einstein field equations in the chosen redshift function ansatz, and it participates in balancing the radial and tangential pressures so that the flare-out condition and geometric requirements can be met. Although in some parameter regimes its direct effect on the stability criteria is subdominant compared to the Casimir and electromagnetic components, it plays a structural role in enabling exact solutions and redistributing the necessary violation of the NEC across multiple matter sectors. Our goal is to study wormholes supported by multiple matter fields. Removing the scalar field would reduce the system to a two-source configuration, changing the exact solutions and eliminating one degree of freedom that currently appears in the integration constants for mathematical ease. In the present context, we have chosen massless scalar field with zero potential and it plays a dominant role in shaping the asymptotic flat WH configurations and associated ECs.

4. Embedding diagrams and quantitative analysis

4.1. Embedding diagrams

Embedding diagrams are essential to study wormhole topology. They help in analyzing how the space-time metric [7] describes a wormhole geometry. To draw embedding diagram, we consider a 2D hyper-surface Σ characterized by the equatorial plane $\theta = \text{constant}$ and $t = \text{constant}$. Therefore, the geometry is constrained to [8]

$$ds_\Sigma^2 = \frac{dr^2}{\left(1 - \frac{b(r)}{r}\right)} + r^2 d\phi^2 \quad (34)$$

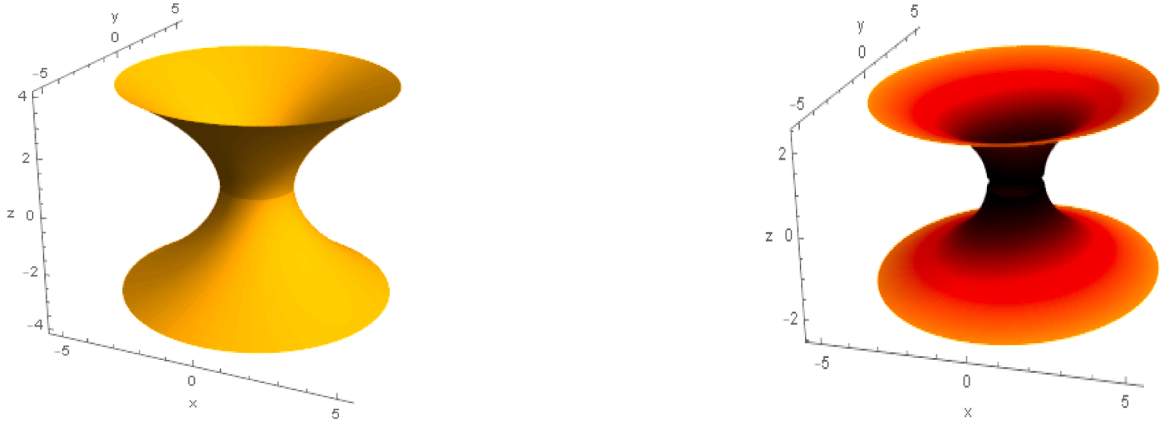


Fig. 3. Embedding diagrams for WH1 and WH2 (left to right).

We may consider this 2D hyper-surface Σ embedded as a rotational surface $z = z(r, \phi)$ into the Euclidean space with line element [8]

$$ds_{\Sigma}^2 = \left(1 + \left(\frac{dz}{dr}\right)^2\right) dr^2 + r^2 d\phi^2 \quad (35)$$

in cylindrical coordinates r, ϕ, z . Comparing Eqs. (34) and (35) we get

$$z(r) = \pm \int_{r_0}^r \sqrt{\frac{b(r)}{r - b(r)}} dr \quad (36)$$

and this $z(r)$ is called the embedding function. For the two WHs, we draw the embedding diagram in Fig. 3.

4.2. Active gravitational mass

The active gravitational mass gives a measure of the effective gravitational influence of the WH structure. The active gravitational mass enclosed within the region from the throat radius r_0 to any arbitrary radius r can be evaluated as [45–48]

$$m_A = 4\pi \int_{r_0}^r \rho u^2 du \quad (37)$$

The expressions for the active gravitational mass for the two WH solutions obtained are

$$m_A|_{WH1} = 4\pi \left(\frac{r_1^2}{\kappa r} - \frac{Cr_0^2}{3\Phi_0^2 r^3} - \frac{Q_0^2}{r} \right) \quad (38)$$

$$m_A|_{WH2} = 4\pi \left(\frac{Q_0^2 + \frac{3r_1^2}{\kappa} - Ce^{-2\Phi_0}}{r} \right) \quad (39)$$

The variation of active gravitational mass has been depicted graphically in Fig. 4. The plots of active gravitational mass shows the matter constituent at the WH throat as well as away from the throat.

4.3. Complexity factor

In 2018, L.Herrera [49] introduced a novel classification for spherically symmetric bodies known as ‘Complexity factor’. It was proposed that a spherically symmetric non-trivial geometry can be classified as complex or non-complex (similarly more complex/ less complex) on the basis of the degree of inhomogeneity and anisotropy of the stress energy tensors. The complexity factor corresponding to a spherically symmetric space-time (1) can be defined as [49]

$$Y_{TF} = 8\pi(p_r - p_t) - \frac{4\pi}{r^3} \int_{r_0}^r u^3 \frac{d\rho(u)}{du} du \quad (40)$$

Clearly, for a homogeneous and isotropic distribution of matter $Y_{TF} = 0$. The expressions for complexity factors for the two obtained WH solutions are given by

$$Y_{TF}|_{WH1} = 8\pi \left(\frac{Cr_0^2}{\Phi_0^2 r^6} - \frac{4Q_0^2}{r^4} - \frac{2r_1^2}{\kappa r^4} \right) \Bigg|_{r_0}^r \quad (41)$$

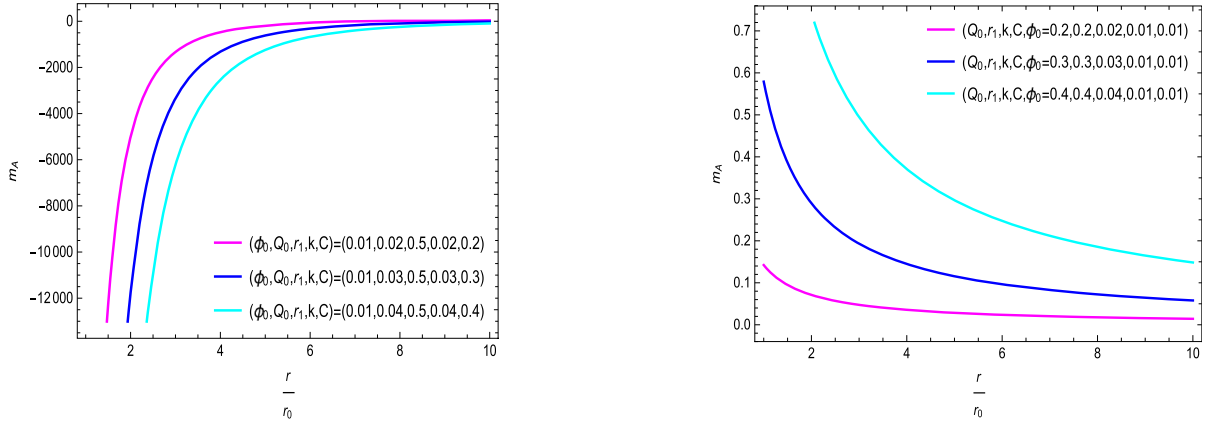


Fig. 4. Variation of active gravitational mass for WH1 and WH2 (left to right).

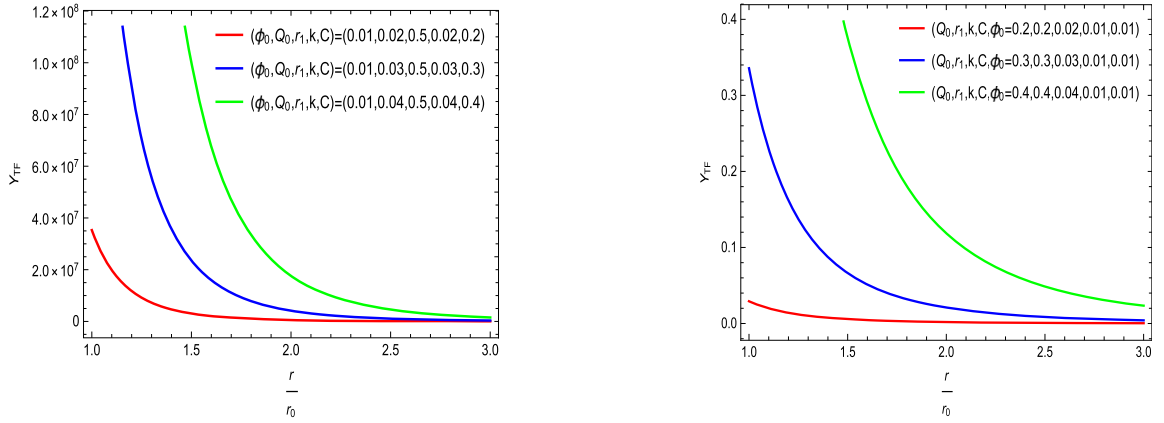


Fig. 5. Variation of complexity factor for WH1 and WH2 (left to right).

$$Y_{TF} \Big|_{WH2} = \frac{16\pi Q_0^2}{r^4} \Big|_{r_0}^r \quad (42)$$

Plots showing the nature of Complexity factor for the obtained WH solutions are depicted in Fig. 5.

Remark. It is observed from the graphs that degree of anisotropy/ inhomogeneity near the WH throat is higher (given by higher Complexity factor) and dies down as we go away from the throat (Complexity factor tends to 0 almost). A high complexity factor means:

1. Strong pressure anisotropy (see first term in definition of complexity factor $(p_r - p_t)$).
2. Large density gradients.(comes from the term under integration)
3. Significant tidal effects.(comes from variable red-shift).

So, higher complexity factor near throat and then a tendency to 0 at large radii implies that the system becomes isotropic and homogeneous at infinity or the spacetime is asymptotically flat/ simple. The plots show that WH2 is less complex than WH1.

4.4. Volume integral quantifier

The quantity of exotic matter essential to sustain the WH geometry can be calculated using Volume Integral Quantifier. It is denoted by I_v and defined by [50]

$$I_v = 2 \int_{r_0}^r 4\pi u^2 (\rho + p_r) du \quad (43)$$

The expressions for the Volume Integral Quantifier for the two obtained WH solutions are given by

$$I_v|_{WH1} = \frac{32\pi}{u} \left(\frac{4r_1^2}{\kappa} - \frac{2Cr_0^2}{3\Phi_0^2 u^2} \right) \Big|_{r_0}^r \quad (44)$$

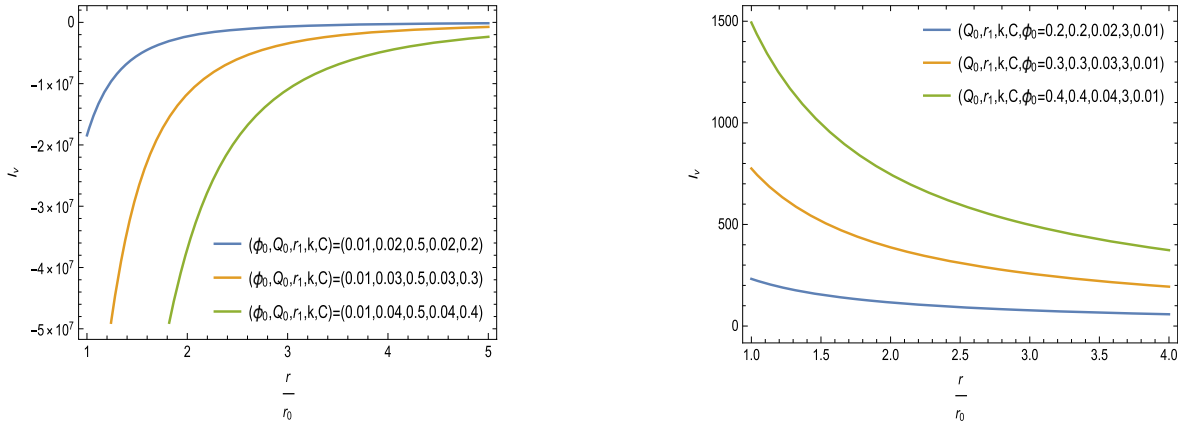


Fig. 6. Variation of volume integral quantifier for WH1 and WH2 (left to right).

$$I_v|_{WH2} = \frac{16\pi}{u} \left(\frac{3r_1^2}{\kappa} - C e^{-2\Phi_0} \right) \Big|_{r_0}^r \quad (45)$$

Plots showing Volume Integral Quantifier for WH 1 and 2 have been shown in Fig. 6. From the plots we have the findings:

$I_v < 0$: high amount of exotic matter required at the throat to support the WH configuration/ high amount of NEC violation is required as seen in case of WH1.

$I_v \geq 0$: less or no amount of exotic matter is required at the throat / lower amount of NEC violation takes place as seen in WH 2.

Remark. More negative is the value of Volume Integral Quantifier, more exotic matter will be required to sustain the WH. Less negative / (closer to zero) value of Volume Integral Quantifier indicates a more physically realistic wormhole, since it requires less exotic matter.

5. Shadows of Casimir WHs: An analytic description

To analyze the shadow of a wormhole (WH), we begin by studying the behavior of light rays governed by null geodesics. The paths of these geodesics follow from the geodesic equation:

$$\frac{dk^\mu}{d\sigma} + \Gamma_{\alpha\beta}^\mu k^\alpha k^\beta = 0, \quad (46)$$

where $k^\mu = \frac{dx^\mu}{d\sigma}$ is the tangent vector to the null geodesics (satisfying $k^\mu k_\mu = 0$), and σ is an affine parameter. Using the Hamilton-Jacobi formalism, one can rewrite this as:

$$\frac{\partial S}{\partial \sigma} = -\frac{1}{2} g^{ab} \frac{\partial S}{\partial x^a} \frac{\partial S}{\partial x^b}, \quad (47)$$

with S being the Jacobi action. Due to the spherical symmetry and static nature of the WH spacetime, conserved quantities emerge: the energy E , angular momentum L , and a Carter-like constant λ . Thus, the separable ansatz for S is:

$$S = \frac{1}{2} m_0^2 \sigma - Et + L\phi + R(r) + T(\theta), \quad (48)$$

where $m_0 = 0$ for photons. Substituting into the Hamilton-Jacobi equation and using the WH metric, one obtains differential equations in r and θ , leading to definitions of effective potentials for radial and angular motion. Introducing the dimensionless impact parameters $\mu = \frac{L}{E}$ and $\nu = \frac{\lambda}{E^2}$, the radial geodesic equation for null particles can be written as:

$$\left(\frac{dr}{d\sigma} \right)^2 = \left(1 - \frac{b(r)}{r} \right) \left(e^{-2\Phi(r)} - \frac{\nu}{r^2} \right), \quad (49)$$

where $b(r)$ is the shape function and $\Phi(r)$ is the redshift function of the WH. This equation takes the form of a classical energy equation:

$$\left(\frac{dr}{d\sigma} \right)^2 + V_{\text{eff}}(r) = 0, \quad (50)$$

with the effective potential defined by:

$$V_{\text{eff}}(r) = \left(1 - \frac{b(r)}{r} \right) \left(\frac{\nu}{r^2} - e^{-2\Phi(r)} \right). \quad (51)$$

To determine the shadow cast by the WH, one considers photon trajectories at the photon sphere radius r_{ph} , where the effective potential satisfies:

$$V_{\text{eff}}(r_{ph}) = 0, \quad \left. \frac{dV_{\text{eff}}}{dr} \right|_{r=r_{ph}} = 0. \quad (52)$$

The radius of the WH shadow as seen by a distant observer is given by:

$$r_{sh} = r_{ph} e^{-\Phi(r_{ph})}. \quad (53)$$

In a coordinate frame where the observer is located at radial infinity and inclination angle θ_0 , the celestial coordinates (X, Y) describing the shadow boundary are:

$$X = -\frac{\mu}{\sin \theta_0}, \quad (54)$$

$$Y = \sqrt{\nu - \frac{\mu^2}{\sin^2 \theta_0}}. \quad (55)$$

In the equatorial plane ($\theta_0 = \pi/2$), these expressions simplify, enabling one to directly trace the shadow outline by plotting (X, Y) . This analytical framework allows the investigation of how various physical parameters influence the shape and size of the WH shadow. In particular, Casimir energy and electric charge can significantly affect the shadow geometry.

Let us consider that the Casimir wormhole (WH) connects two distinct asymptotically flat regions of spacetime. Suppose one of these regions contains a luminous source, while the other is devoid of any nearby light emission near the WH throat. In this situation, one may have two types of orbits in the first region namely [55]

(a) photons that traverse the wormhole by passing through its throat and (b) photons that are deflected and escape to infinity. To a distant observer situated in the bright region, only the deflected photons are visible. Photons that pass through the wormhole are not detected, leading to the formation of a dark patch—known as the wormhole's shadow—against the luminous background. For such scattering to occur, the radial trajectory of photons must exhibit a turning point, mathematically represented by $\frac{dr}{d\sigma} = 0$ i.e., $V_{\text{eff}} = 0$ in order to have scattering phenomena. The boundary separating plunge and escape trajectories is defined by a critical orbit, located at the maximum of the effective potential. This unstable circular orbit, also called the photon sphere, is sensitive to perturbations: even a slight deviation can cause the photon to either fall into the wormhole or escape. Therefore, this critical behavior is captured by the condition $V_{\text{eff}} = 0 = \frac{dV_{\text{eff}}}{dr}$ and $\frac{d^2V_{\text{eff}}}{dr^2} \leq 0$. So for the present Casimir WH configuration, the radius of the photon sphere r_{ph} which locates the apparent image of the photon rings satisfies [51–55]

$$r\phi'(r) = 1 \quad (56)$$

i.e.,

$$e^{\phi(r)} = \phi_0 \left(\frac{r}{r_0} \right) \quad (57)$$

The photon radius, r_{ph} is the largest positive real root of the equation $r\phi'(r) = 1$ and the impact parameter ν identifies the shadow radius as

$$r_{sh} = \sqrt{\nu} = r e^{-\phi(r)}|_{r=r_{ph}} \quad (58)$$

where r_{sh} denotes the radius of the shadow. Conveniently, we assume that the observer is situated far away from the WH so that radius of the WH shadow can be expressed as [55]

$$r_{sh} = r_{ph} e^{-\phi(r_{ph})} = \sqrt{\mathcal{X}^2 + \mathcal{Y}^2} \quad (59)$$

where in the observer's frame $(\mathcal{X}, \mathcal{Y})$ are identified as the celestial coordinates. Let, r_0 is the distance of the observer from the WH and θ_0 is the observer's angular coordinate (or inclination angle) i.e., (r_0, θ_0) is the location of the observer, then $(\mathcal{X}, \mathcal{Y})$ can identify the boundary curves of the WH shadow (i.e., the apparent shape of the shadow) and are related to the observer's coordinates (r_0, θ_0) as [51–55]

$$\mathcal{X} = \lim_{r_0 \rightarrow \infty} (-r_0^2 \sin \theta_0) \frac{d\phi}{dr} \quad (60)$$

$$\mathcal{Y} = \lim_{r_0 \rightarrow \infty} \left(r_0^2 \frac{d\theta}{dr} \right) \quad (61)$$

i.e.,

$$\mathcal{X} = -\frac{\mu}{\sin \theta_0} \quad (62)$$

$$\mathcal{Y} = \sqrt{\nu - \mu^2 / \sin^2 \theta_0} \quad (63)$$

From the above interrelations among the celestial coordinates and the impact parameters one may form the shadow of the Casimir WH. To draw shadow, one has to plot $\mathcal{X}$ vs $\mathcal{Y}$ to identify the shadow's boundary and for simplicity one may choose the equatorial plane

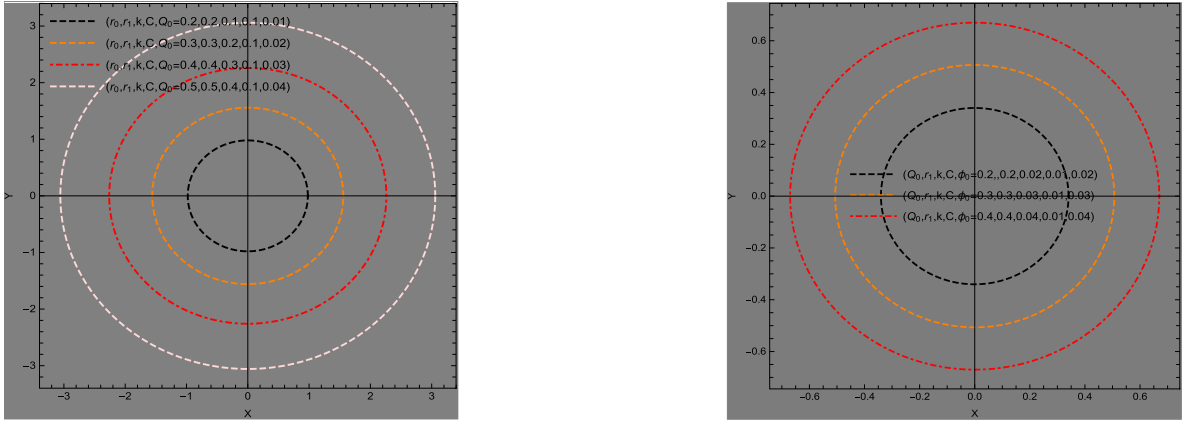


Fig. 7. Shadows of WH1 and WH2 (left to right).

$\theta_0 = \frac{\pi}{2}$. For the WH1 case, the “observer at infinity” in our figures/calculations should be understood as located in the asymptotically-flat region of the matched exterior spacetime. The shadows of the Casimir WHs 1 and 2 have been depicted graphically in Fig. 7.

Remark. More is the Casimir effect, greater is the size of the shadow for WH1 and WH2.

6. Quasinormal modes and grey body factors of Casimir WHs

In the field of gravitational wave (GW) astronomy, growing attention has been directed toward the investigation of quasinormal modes (QNMs) and greybody factors (GBFs) in the context of wormholes (WHs) [36,37]. These quantities characterize how the wormhole spacetime responds to perturbations and serve as potential observational indicators. QNMs represent damped oscillatory modes intrinsic to the geometry of the WH, determined by the spacetime configuration and associated boundary conditions. Conversely, GBFs quantify the likelihood that perturbative waves can transmit through the wormhole’s effective potential barrier.

Together, QNMs and GBFs are considered promising tools for distinguishing wormholes from black holes (BHs) in observational settings, particularly within the framework of GW studies. When small perturbations are introduced into the wormhole background—or equivalently, when test fields evolve in this spacetime—their dynamics are governed by a Schrödinger-like wave equation of the form: [36,37]

$$\left(\frac{d^2}{dr_T^2} + \omega^2 - V(r_T) \right) \Psi(r_T) = 0 \quad (64)$$

Here, ω represents the frequency of oscillation, V is the effective potential that depends on the WH geometry and the tortoise coordinate r_T has the expression

$$r_T(r) = \int \frac{dr}{e^{\Phi} \sqrt{1 - \frac{b(r)}{r}}} \quad (65)$$

The expression for the effective potential is given by [8]

$$V(r) = e^{2\Phi} \left(\frac{l(l+1)}{r^2} - \frac{(rb' - b)}{2r^3} + \frac{\Phi'(r)}{r} \left(1 - \frac{b(r)}{r} \right) \right) \quad (66)$$

for massless scalar field and

$$V(r) = e^{2\Phi} \frac{l(l+1)}{r^2} \quad (67)$$

for electromagnetic field. However, in the present context we will analyze the effective potential for massless scalar field and not the electromagnetic field. Here l (the angular momentum quantum no.) represents the multipole number. The QNMs ($Re(\omega) + iIm(\omega)$) of WHs are the solutions of the above wave Eq. (64) with the purely outgoing boundary conditions at infinity i.e.,

$$\Psi(r_T) \approx \Psi_I e^{\pm i\omega r_T}, \quad r_T \rightarrow \pm\infty \quad (68)$$

It is to be noted that, the QNM spectra gives information regarding the evolution of the perturbed field. The real part of ω identifies the oscillation of the signal while the imaginary part measures the damping factor for the energy loss of gravitational radiation. Precisely, if $\Psi(r_T) \approx \Psi_I e^{i\omega r_T}$, $r_T \rightarrow \pm\infty$ then $Im(\omega) \leq 0$ gives unstable perturbations with exponential growth of the perturbed field and $Im(\omega) > 0$, implies stability of the perturbed field. Reversely, if $\Psi(r_T) \approx \Psi_I e^{-i\omega r_T}$, $r_T \rightarrow \pm\infty$ then $Im(\omega) < 0$ gives stable perturbations and

$Im(\omega) > 0$ gives unstable perturbations. On the other hand, due to Schrödinger like nature of wave Eq. (64) the WKB approximation is preferable in this arena. Thus, using WKB approach the QNM frequency ω near the WH throat takes the form [8]

$$\omega = \frac{e^{\Phi(r_0)}}{r_0} \left(l + \frac{1}{2} \right) - i \left(n + \frac{1}{2} \right) \frac{e^{\Phi(r_0)}}{\sqrt{2}r_0} + O(l^{-1}) \quad (69)$$

Hence the real part has a relation with the shadow radius as

$$Re(\omega) = \frac{1}{r_{sh}} \left(l + \frac{1}{2} \right) \quad (70)$$

The expression above indicates that the shadow radius is influenced not only by the location of the wormhole throat but also by the presence of the outer photon sphere, with $r_{ph} > r_0$. This relationship holds exactly in the eikonal limit, where the angular momentum quantum number l is large, and remains a good approximation even for relatively small l . In this regime, the behavior of the scalar field closely mirrors that of the geometric optics approximation.

Furthermore, since the functions $b(r)$ (shape function) and $\phi(r)$ (redshift function) are continuous, they can be expanded in a Taylor series around the throat $r = r_0$, given by [8]

$$b(r) = b_0 + b_1(r - r_0) + b_2(r - r_0)^2 + \dots \quad (71)$$

$$\Phi(r) = \Phi_0 + \Phi_1(r - r_0) + \Phi_2(r - r_0)^2 + \dots \quad (72)$$

It is worth mentioning that in asymptotically flat spacetimes, incorporating a pre-factor may be necessary to capture the correct behavior at infinity. However, since our analysis is focused near the wormhole throat, such a pre-factor is not required in this context.

For large values of the angular momentum l , the quasinormal mode (QNM) frequencies can be approximated analytically using the WKB method. The expression for the QNM frequencies at the k -th order of the WKB approximation is given by [8]:

$$i \frac{\omega^2 - V_m}{\sqrt{-2V_m''}} - \sum_{i=2}^k \Gamma_i = n + \frac{1}{2} \quad (73)$$

Here, V_m represents the maximum of the potential, V_m'' is its second order derivative w.r.t the tortoise coordinate, all higher order corrections are contained in Γ_i 's and k stands for the order of the WKB approximation. It is crucial to note that large n does not give a good approximation for the QNM frequencies. In fact, the value of k for best value of QNM frequency is not unique it depends on (n, l) . Further, one can relate the shape of WH with the imaginary part of QNM as

$$Im(\omega) = \frac{\sqrt{(b_1 - 1)(b_0\Phi_1 - 1)}}{\sqrt{2}r_{sh}} \quad (74)$$

Hence, the imaginary component of the quasinormal frequency which dictates how quickly the wave amplitude decays—is also influenced by the shadow radius. Specifically, a larger shadow radius corresponds to a slower rate of damping. Also, $b'(r = b_0) = 1$ gives $Im(\omega) = 0$ i.e., one has pure real modes as the standing waves of an oscillating string with fixed ends vanishing at the throat. To study wave scattering near the wormhole, one must consider how waves interact with the effective potential barrier surrounding the wormhole—or more generally, any compact object. This interaction results in a portion of the wave being reflected and another portion transmitted through the barrier. Greybody factors (GBFs) account for both of these outcomes. Importantly, GBFs are independent of the wave's origin—whether it comes from the asymptotic region of the second universe or from spatial infinity in the same universe. Accordingly, the boundary conditions in this context are modified as follows:

$$\begin{aligned} \Psi &= e^{-i\Omega r_T} + R e^{i\Omega r_T}, \quad r_T \rightarrow +\infty \\ \Psi &= T e^{-i\Omega r_T}, \quad r_T \rightarrow -\infty \end{aligned} \quad (75)$$

where R and T are the reflection and transmission coefficients. Here, the frequency Ω is real and continuous for scattering phenomena and is distinct from the complex and discrete QNM frequency. The transmission coefficient T is usually termed as the GBF as it measures the fraction of the wave which traverses the potential barrier and as a result it contributes to the emission of radiation from the compact objects like a BH or a WH. Therefore, one may write the GBF corresponding to the angular momentum number l as [8]

$$\Gamma_l(\Omega) = |T|^2 = 1 - |R|^2 \quad (76)$$

Using WKB expression for the GBF one has [8]

$$\Gamma_l(\Omega) = \frac{1}{1 + e^{2\pi i k}} \quad (77)$$

Now, the effective potential can be expanded in powers of l as

$$V(r_T) = l^2 V_0(r_T) + l V_1(r_T) + l^{-1} V_2(r_T) + \dots \quad (78)$$

and as a result the eikonal approximation can be obtained from the first order WKB approximation as follows

$$\Omega = l \sqrt{V_{00}} - i k \sqrt{\frac{-V_{00}''}{2V_{00}}} + O(l^{-1}) \quad (79)$$

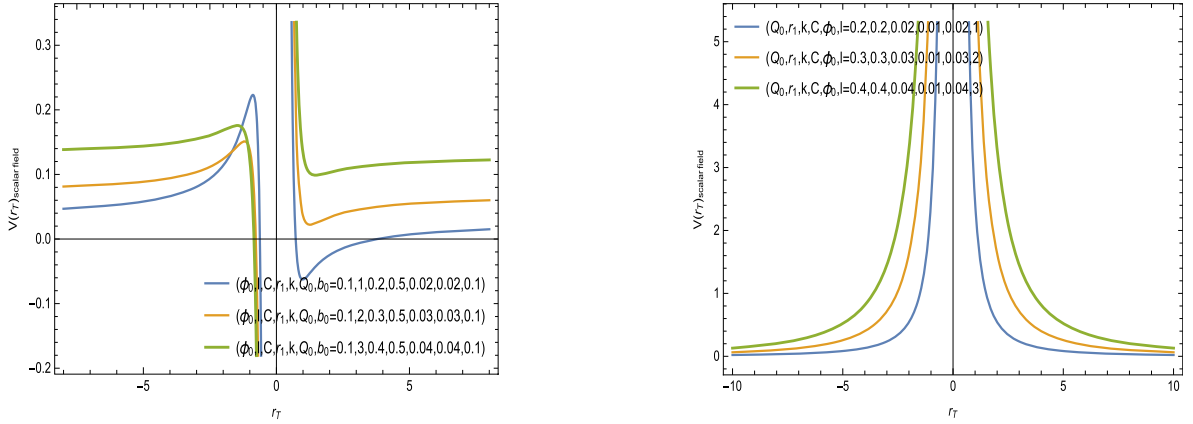


Fig. 8. Effective potential barrier corresponding to massless scalar field as function of tortoise coordinate for WH1 and WH2 (left to right).

where V_{00} is the value of the function $V_0(r_T)$ in its maximum and V''_{00} is value of second derivative of $V_0(r_T)$. As a consequence, k can be expressed as a function of the real frequency Ω

$$-ik = \frac{\Omega^2 - l^2 V_{00}}{l \sqrt{-2V_{00}}} + O(l^{-1}) = -\frac{\Omega^2 - \text{Re}(\omega_0)^2}{4 \text{Re}(\omega_0) \text{Im}(\omega_0)} + O(l^{-1}) \quad (80)$$

Hence, one may relate the transmission coefficient to the GBFs $\Gamma_l(\Omega)$ with the fundamental mode ω_0 as [8]

$$\Gamma_l(\Omega) = |T|^2 = \left(1 + e^{\frac{2\pi}{4 \text{Re}(\omega_0) \text{Im}(\omega_0)} (\Omega^2 - \text{Re}(\omega_0)^2)} \right) \quad (81)$$

It is to be noted that the above relation is exact in the eikonal limit $l \rightarrow \pm\infty$ and an approximate one for small l . To find the interrelation between radius of the shadow with GBF we consider Eqs. (71)–(80) to get

$$V_{00} = e^{2\Phi_0} \left(\frac{4l(l+1)}{2r_0^4} ((2r_0^2(\Phi_2 + \Phi_1^2)) - 1) + \frac{b_2}{r_0^3} (1 - 2\Phi_1 r_0) - \frac{3b_3}{r_0^2} \right)$$

In the present Casimir WHs the expression for the effective potential barriers are given by

$$V(r_T)|_{WH1} = \Phi_0^2 \left(l(l+1) - \frac{\left(1 - \frac{1}{r_T}\right)}{2r_T} \left(\frac{Cr_0^2}{2\Phi_0^2 r_T^3} - \frac{r_1^2}{\kappa r_T} + \frac{Q_0^2}{r_T} \right) \right) + \left(1 - \frac{b_0}{r_T} + \frac{Cr_0^2}{6\Phi_0^2 r_T^4} - \frac{r_1^2}{\kappa r_T^2} + \frac{Q_0^2}{r_T^2} \right)$$

$$V(r_T)|_{WH2} = e^{2\Phi_0} \left(\frac{l(l+1)}{r_T^2} + \frac{\sqrt{3r_1^2 + \kappa Q_0^2 - \kappa C e^{-2\Phi_0}}}{r_T^4} \right)$$

The plots showing the effective potential barrier for WH 1 and 2 are shown in Fig. 8.

The expressions for real part and imaginary part of QNM frequency (both arbitrary branch and principal branch/fundamental mode/lowest overtone corresponding to $n = 0$) corresponding to the two WH solutions are obtained as

$$\text{Re}(\omega)|_{WH1} = \frac{\left(l + \frac{1}{2}\right) \sqrt{C\kappa}}{r_0 \sqrt{6(r_1^2 - \kappa Q_0^2)}}$$

$$\text{Re}(\omega)|_{WH2} = \frac{2e^{\Phi_0} \left(l + \frac{1}{2}\right)}{\sqrt{3r_1^2 + \kappa Q_0^2 - \kappa C e^{-2\Phi_0}}}$$

$$\text{Im}(\omega)|_{WH1} = 0, \quad \text{Im}(\omega)|_{WH2} = \frac{\sqrt{1 + \frac{1}{\sqrt{3r_1^2 + \kappa Q_0^2 - \kappa C e^{-2\Phi_0}}}}}{e^{-\Phi_0} \sqrt{3r_1^2 + \kappa Q_0^2 - \kappa C e^{-2\Phi_0}}}$$

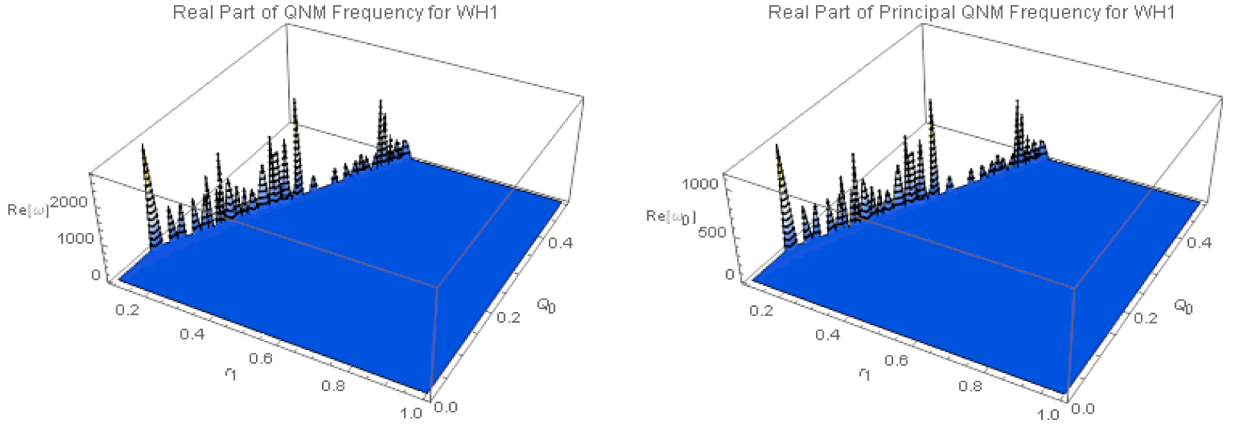


Fig. 9. Variation of QNM spectra with charge and Casimir energy for WH1.

$$\begin{aligned} \text{Re}(\omega_0)|_{WH1} &= \frac{l\sqrt{C\kappa}}{r_0\sqrt{6(r_1^2 - \kappa Q_0^2)}}, & \text{Re}(\omega_0)|_{WH2} &= \frac{2le^{\Phi_0}}{\sqrt{3r_1^2 + \kappa Q_0^2 - \kappa C e^{-2\Phi_0}}} \\ \text{Im}(\omega_0)|_{WH1,2} &= 0 \end{aligned} \quad (82)$$

The QNM spectra for WHs 1 and 2 are shown in Figs. 9 and 10. In Fig. 8, we investigate the behavior of the effective potential $V(r_T)$ for scalar field perturbations as a function of the tortoise coordinate r_T , focusing on how the potential profile is influenced by key parameters including the red-shift function at the throat Φ_0 , electric charge Q_0 , massless scalar field coupling constant C , and the multi-pole number l . These potentials play a central role in determining the scattering, stability, and quasi-normal mode structure of the underlying spacetime. Φ_0 controls gravitational red-shift. Higher Φ_0 increases barrier height and sharpness prominent in WH2. Charge Q_0 introduces symmetric repulsion near the throat and help in raising the effective potential prominent in WHs 1 and 2. In all the plots of Fig. 8, it is clear that the effective potential encounters barrier near $r_T = 0$. Coupling constant C affects potential asymmetry and depth as evident in WH 1. Since C remains unchanged in case of WH 2 so there is no potential asymmetry and the effective potential barrier is symmetric about $r_T = 0$. Finally, l the angular momentum number raises centrifugal barrier, enhancing steepness of the potential.

Fig. 9 shows that the real part of the QNM frequencies for WH1 shows sharp peak with the Casimir energy parameter r_1 indicating that stronger quantum vacuum support leads to higher-frequency oscillations. On the other hand, presence of electric charge Q_0 further amplifies this effect but more smoothly (peaks are not so sharp but smooth). The vanishing of real frequencies at high r_1 and low Q_0 indicates a transition to a non-oscillatory regime, which might serve as a distinct observational feature of Casimir-driven wormholes. The comparison between the left and right figures reveal that the principal QNM mode (right figure) exhibits a qualitatively similar but quantitatively damped profile compared to the full spectrum (left figure). These results highlight that both Casimir energy and electric charge have significant influence on the QNM spectrum of WH 1. In particular, regions of the parameter space with high Q_0 and low r_1 yield high-frequency modes, which could be critical for potential observational signatures, such as gravitational wave echoes or deviations from standard black hole ring-down signals.

First Fig. 10 on the left shows that larger Casimir parameters or wormhole charge reduce the oscillation frequency of perturbations. As both r_1 and Q_0 increase, the real part of the frequency decreases. Second Fig. 10 on the right shows the variation of fundamental mode w.r.t the Casimir parameter and charge. The surface shows similar behavior as the full QNM spectrum's real part, suggesting that the fundamental mode dominates the dynamics. Again, increasing r_1 or Q_0 lowers the frequency. The resemblance to the first plot indicates the fundamental mode or typically the lowest overtone contributes significantly to the total QNM spectra in WH2. Third figure of Fig. 10 shows that greater is the Casimir effect or effect of charge, lesser is the damping as $\text{Im}(\omega)$ represents damping. The perturbations are short-lived for small r_1 and Q_0 , and become longer-lived as these parameters increase.

From the expression for GBF in Eq. (81) we find that $\Gamma_l(\omega) \rightarrow 1$ if $\Omega^2 < \text{Re}^2(\omega_0)$. This is because, $\text{Im}(\omega_0) = 0$. Therefore, for the two WH solutions we have the following constraints to have a finite value of the GBF tending to unity.

$$\begin{aligned} \Omega^2|_{WH1} &< \frac{l^2 C \kappa}{6r_0^2(r_1^2 - \kappa Q_0^2)} \\ \Omega^2|_{WH2} &< \frac{4l^2 e^{2\Phi_0}}{3r_1^2 + \kappa Q_0^2 - \kappa C e^{-2\Phi_0}} \end{aligned}$$

If a WH has GBF close to 1 it implies that the WH is nearly transparent to radiation/ perturbations. GBF actually quantifies the probability that radiation will transmit through potential barrier surrounding a compact object. GBF unity implies perfect transmission i.e., incoming radiation is not reflected and can propagate to the WH throat to the other side (or escape into infinity if it is an outgoing wave). Thus, for the two Casimir WHs, we observe that WH's geometry allows waves to pass through minimal distortion. The GBF

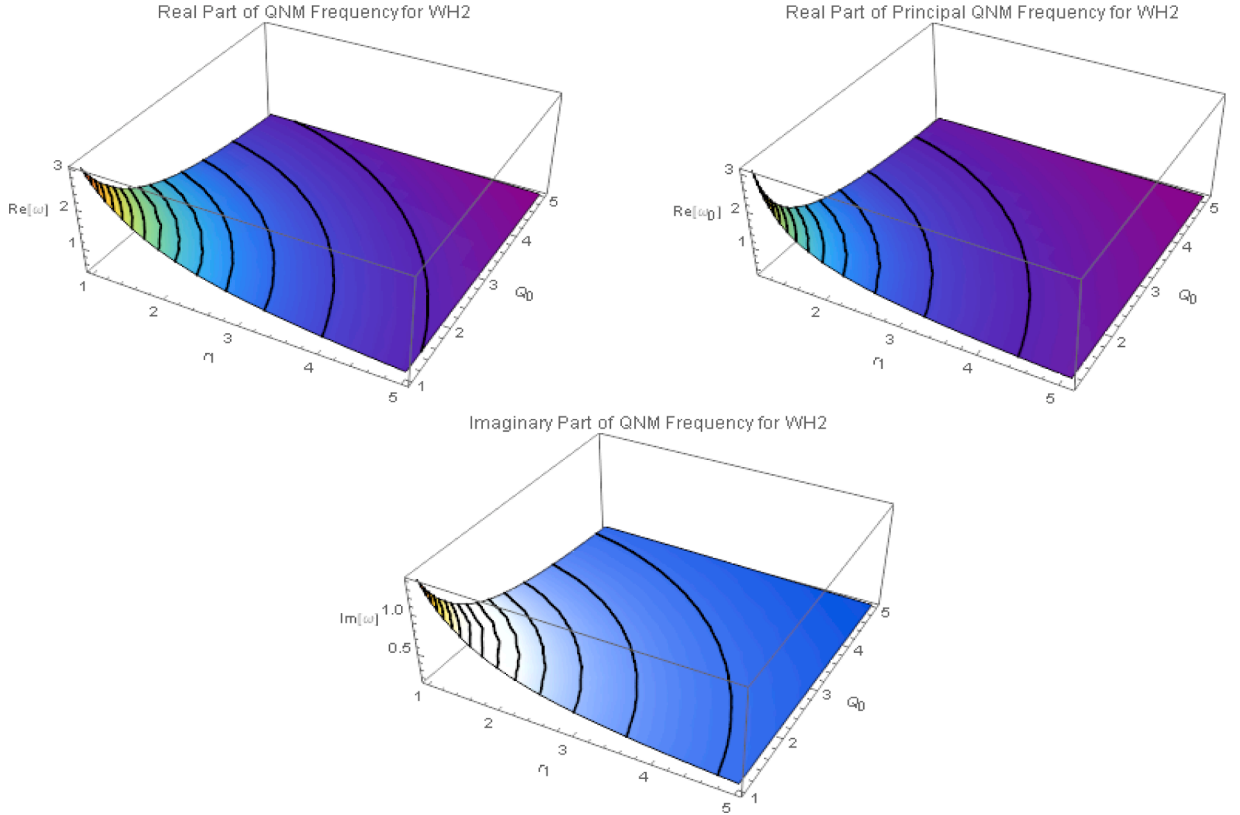


Fig. 10. Variation of QNM spectra with charge and Casimir energy for WH2.

analysis supports traversability and non-trapping geometry. If a gravity wave detector like LIGO, LISA observe an object emitting radiation that appears only slightly distorted from a thermal / black-body spectra, it might indicate GBF close to 1, hinting at a non black hole exotic object like the WHs. Similar results apply here for Casimir WHs.

7. Conclusion

In this paper, we have explored and constructed novel TWH geometries supported by the Casimir effect in association with classical electric fields and a massless scalar field with vanishing potential. The objective was to investigate whether quantum vacuum fluctuations, specifically the experimentally verified Casimir effect, could realistically support the exotic matter necessary for wormhole stability, thereby bridging GR and QFT.

Two distinct wormhole solutions were derived under different assumptions on the red-shift function. Each solution was minutely analyzed with respect to traversability condition, study of classical energy conditions and physical features such as asymptotic flatness and the flaring-out condition at the throat. Remarkably, the two wormholes maintained the geometric structure necessary for traversability, but differed in the matter content. For WH1, it is observed that with Casimir effect, massless scalar field with vanishing potential and electric field it is possible to get asymptotic flat TWH satisfying flaring out criteria and the combined effects characterize a WH configuration where classical ECs may hold if the effect of massless scalar field with zero potential coming via C and the electromagnetic field via Q_0 both are dominant over the Casimir source via r_1 . While for WH 2, the SEC and NEC hold and DEC is violated if the effect of Casimir energy is minimal via r_1 and the effect of massless scalar field via C is dominant.

Further quantitative studies were carried out by using embedding diagrams, active gravitational mass, complexity factors, and the volume integral quantifier. These tools enabled a deeper understanding of the matter distribution, gravity profile and structural simplicity of the wormholes.

Shadows of WH1 and WH2 were notably influenced by Casimir energy, growing in size with stronger quantum effects. These results emphasize the potential of shadow imaging (as with Event Horizon Telescope observations) as an insight for identifying and distinguishing wormhole geometries.

In addition, the study of QNMs and GBFs, the key signatures in gravity wave astronomy, established a compelling interrelation between wormhole stability and its observational footprint. The real and imaginary parts of the QNM frequencies were directly related to the wormhole shadow radius and throat geometry. GBFs computed via WKB approximations, provided insights into radiation

transmission characteristics of these Casimir WHs. Notably, the link between shadow radius and the damping rate of QNMs presents a unified theoretical and observational framework to distinguish TWHs from BHs.

Therefore, the present study confirms that Casimir-driven TWHs are not only theoretically consistent within the framework of GR and QFT, but they may also exhibit observable features like shadows, quasinormal ringing, and grey body spectra. Future extensions of this work might explore dynamical stability under perturbations, inclusion of accretion effects, and more robust constraints using gravity wave data and shadow imaging by EHT datasets.

CRediT authorship contribution statement

Madhukrishna Chakraborty: Writing – original draft, Visualization, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Subenoy Chakraborty:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Methodology, Investigation, Formal analysis, Conceptualization.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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